Proof and Induction WS 3

1. Use a direct proof to prove each of the following statements.
    a. The sum of the squares of five consecutive integers is divisible by 5.
    b. The product of two rational numbers is rational.
2. Use a contrapositive proof to prove this statement "*Let n be an integer. If $n^3 + 5$ is odd, then n is even*."
3. Use a proof by contradiction to prove this statement "*The sum of a rational and an irrational number is irrational*".
4. Prove the following logical equivalences. Let n be a positive integer. n is even if and only if $13n + 4$ is even.
5. Prove that if $a^2 - 2a + 7$ is even, then a is odd.
6. Use a proof by contradiction to show that there is no rational solution to the equation $x^3 + x + 1 = 0$. As a hint, start by supposing, for a contradiction that $r = \frac{p}{q}$ is a rational solution to the equation, where p, q are integers with no common factor other than 1 and with $q \neq 0$. Then consider what would happen if both p and q were odd, or if one of them was even and the other odd.
7. Write the contrapositive of each statement.
    a. $A \Rightarrow B$
    b. $\neg P \Rightarrow Q$
    c. $N \Rightarrow \neg M$
    d. $\neg B \Rightarrow \neg F$
    e. If the boy has red hair, then he has blue eyes.
    f. If the country is rich, then the citizens has money.
    g. If a quadrilateral is a kite, then the adjacent sides are equal in length.
    h. If $x = y$, then $x^2 = y^2$
    i. If $a \in \mathbb{N}$, then $a \in \mathbb{Z}$
8. The contrapositive of "If it barks it's a dog" is
    (A) "If it doesn't bark, it isn't a dog"
    (B) "If it's a dog it will bark"
    (C) "If it doesn't bark it's a dog"
    (D) "If it isn't a dog it doesn't bark"
9. Which of the following is true?
    (A) $\forall\, a \in Z^+, \exists b \in Z^+ : b = a^3$
    (B) $\forall\, a \in Z^+, \exists b \in Z^+ : b = \sqrt[3]{a}$
    (C) $\forall\, a \in Z^+, \exists b \in Z^+ : a = b^3$
    (D) $\forall\, a, \exists b \in Z^+ : b = a^3$
10. a. Prove by contraposition that if $n^2 + 6n$ is even, then n is even.
    b. Prove by contradiction that $\log_3 11$ is irrational.
    c. Prove by induction that $3^n \geq n^2$ for all positive integers $n \geq 1$.
    d. Prove for all integers x, y that if $10x + y$ is divisible by 17, then $3y - 4x$ is also divisible by 17.
11. Given that x and y are real numbers, which of the following is a true statement:
    (A) $\forall y(\exists x : |x| = y)$
    (B) $\forall y(\exists x : |x| < y)$
    (C) $\forall y(\exists x : |x| > y)$
    (D) $\forall y(\exists x : |x| = -y)$
12. If $u_1 = 7$ and $u_n = 2u_{n-1} - 1$ for $n \geq 2$, show that $u_n = 3.2^n + 1$ for $n \geq 1$.
13. If $u_1 = 2, u_2 = 16$ and $u_n = 8u_{n-1} - 15u_{n-2}$ for $n \geq 3$, show that u_n is divisible by 64 for $n \geq 1$.
14. If $u_n = 3^n - 2n - 1$, show that $u_{n+1} = 3u_n + 4n$, and hence show that $u_n > 0$ for $n \geq 2$.

15.

a. Show that $(1+x)^n - 1$ is divisible by x for $n \geq 1$.

b. Show that $(1+x)^n - nx - 1$ is divisible by x^2 for $n \geq 2$.

16. If $u_1 = 5, u_2 = 11$ and $u_n = 4u_{n-1} - 3u_{n-2}$ for $n \geq 3$, show that $u_n = 2 + 3^n$ for $n \geq 1$.

17. Prove by mathematical induction that for all positive integers n, $\tan x + 2\tan 2x + 4\tan 4x + \cdots + 2^{n-1}\tan(2^{n-1}x) = \cot x - 2^n \cot(2^n x)$.

18. Show that for $n \geq 1$, $1\ln\frac{2}{1} + 2\ln\frac{3}{2} + \cdots + n\ln\left(\frac{n+1}{n}\right) = \ln\left(\frac{(n+!)^n}{n!}\right)$.

19. If $u_1 = 1$ and $u_n = \sqrt{3 + 2u_{n-1}}$ for $n \geq 2$

a. Show that $u_n < 3$ for $n \geq 1$

b. Deduce that $u_{n+1} > u_n$ for $n \geq 1$.

20. The number $f_n = 2^{2^n} + 1$ for $n \in \mathbb{Z}^+$, where 2^{2^n} is 2 raised to the power of 2^n

Prove, using mathematical induction, that for all $n \in \mathbb{Z}^+$, $f_0 f_1 f_2 \dots f_{n-1} = f_n - 2$

21. The Fibonacci sequence is defined as follows: $T_1 = T_2 = 1$ and $T_n = T_{n+1} + T_{n+2}$ for $n \geq 3$.

Prove by induction that:

a. $T_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$,

b. $T_n < \left(\frac{7}{4}\right)^n$

c. T_{4n} is divisible by 3.

22. Show that for $n \geq 1, n! \geq 2^{n-1}$. Deduce that $\frac{1}{(1!)^2} + \frac{1}{(2!)^2} + \cdots + \frac{1}{(n!)^2} \leq \frac{4}{3}\left(1 - \frac{1}{4^n}\right)$

23. Chris believes that if $x + \frac{1}{x}$ equals an integer, say N, then $x^n + \frac{1}{x^n}$ must also be an integer, for all positive integers n.

a. Prove that this is true for $n = 1$ and $n = 2$

b. Hence use strong induction to prove that if $x + \frac{1}{x}$ is an integer then $x^n + \frac{1}{x^n}$ is an integer, for all integers $n \geq 1$.

24. Prove by mathematical induction that $1 + 2q + 3q^2 + \cdots + nq^{n-1} = \frac{1-(n+1)q^n + nq^{n+1}}{(1-q)^2}$ for all positive integral n.

25. Assuming the triangular inequality $|b + c| \leq |b| + |c|$, prove by induction that: $|x_1 + x_2 + x_3 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$

26. If $u_1 = 1$ and $u_n = \sqrt{2u_{n-1}}$ for $n \geq 2$

a. Show that $u_n < 2$ for $n \geq 1$

27. Deduce that $u_{n+1} > u_n$ for $n \geq 1$

Answer

7. a. $\neg B \Rightarrow \neg A$; b. $\neg Q \Rightarrow P$; c. $M \Rightarrow \neg N$; d. $F \Rightarrow B$; e. If the boy does not have blue eyes then he does not have red hair; f. If the citizen don't have money then the country is not rich; g. If a quadrilateral does not have adjacent sides equal in length, then it is not a kite; h. If $x^2 \neq y^2$ then $x \neq y$; i. If $a \notin \mathbb{Z}$ then $a \notin \mathbb{N}$

8. D 9. A 11. C